



Linux & Bash

Introduction

Introduction to Linux

- ❖ An **Operating System** is a system software that manages resources in a computer system to provide services and functionalities. E.g. UNIX
- ❖ In 1969, **Ken Thompson** and **Dennis Ritchie** developed **UNIX** at Bell Labs
- ❖ In 1983, Richard Stallman started the **GNU**(GNU is **Not Unix**) project to create a free and open source UNIX-like operating system
- ❖ in 1991, **Linus Torvalds** released a **kernel** named **Linux** as a free software.
 - A kernel is the core and the main part of an operating system
- ❖ In 1992, Linux was released under GNU General Public License and Linux and GNU developers started to integrate Linux and GNU components to create a fully functional free and open source OS named **GNU/Linux**

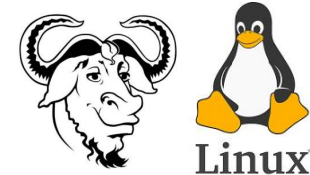
Application

Operating System

Hardware

A computer system

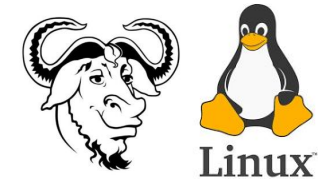
 [History of Linux](#)



Introduction to Linux - Features

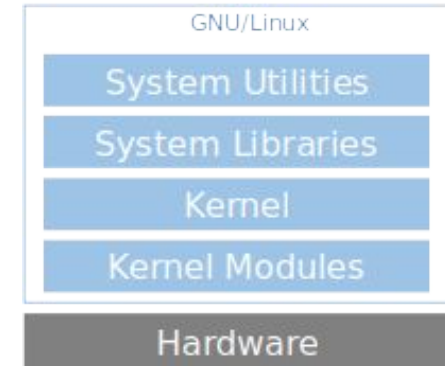
❖ Some features of Linux

- Freely available to use
- Open source; possible to modify and distribute it without any permissions.
 - There are a lot of different Linux based OS distributions. E.g. Debian, Ubuntu, Kali, and etc.
- Portable, flexible and stable; it can be run on different type of computer systems
 - E.g. Embedded systems, mobiles, computers and etc.
- Multi-user and multi-process
- High performance and secure; by providing different mechanisms.
 - E.g. Authentication, file access control, password protection, data encryption and etc.
- Compatible with a lot of of file systems by providing a hierarchical virtual file system.
 - E.g. FAT, NTFS, ext, and etc.
- Has a large community support.

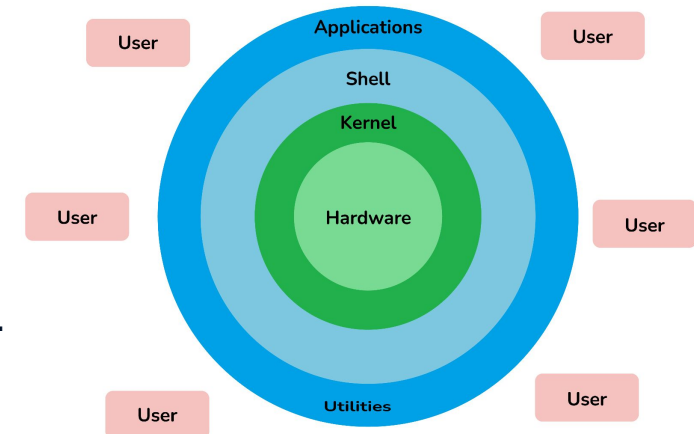


Introduction to Linux - Architecture

- ❖ Why can we power different devices using a plug and a socket?
- ❖ Linux is a monolithic kernel with a modular design
 - **Kernel Modules** are used to abstract hardware
 - They can be loaded and unloaded into the kernel at runtime
 - **Kernel** is the core of the OS. It is hardware independent and provides a standard system call interface for the upper layer
 - **System libraries** are used to get access to the Kernel features and functionalities
 - **System Utilities** to perform system-level tasks and provide functionalities of the operating system to the user.
 - **Shell** is an interface between user and kernel



[Architecture and features](#)



Introduction to Linux

- ❖ Directory structure in Linux is defined by [Filesystem Hierarchy Standard](#).
- ❖ **Shell** is a program that takes commands and passes them to the OS to perform.
 - A powerful command line interface to the system and a scripting language interpreter.
 - A command language is an interpreted language for job control in computing. E.g. shell, batch
- ❖ There are different types of shell. E.g. zsh, ksh, **bash**, and etc.
- ❖ **Bash**(**B**ourne **A**gain **S**hell) is the GNU Project shell.
 - It offers both [scripting](#) and user interaction.
- ❖ A **Terminal Emulator** is a graphical command line interface to interact with the shell.
 - Multi-user and Multi-session
 - Each session is a separate environment

Introduction to Linux - Some bash commands

- ❖ **clear**: to clear the terminal; **exit**: to terminate a session.
- ❖ To get help for bash commands you can use **--help**. E.g. `ls --help`
- ❖ It is possible to use [wildcards to make patterns](#).
- ❖ **ls** (list storage): lists files and subdirectories in the current or give directories.
 - There are some options for this command. For example:
 - **ls [dir_path]**: lists files and subdirectories in dir_path
 - **ls -a**: lists all the files and subdirectories (even hidden) in the current directory
 - **ls -l**: long list of all the files and subdirectories with details
- ❖ **cd <dir_name>**: Changes the working directory to **dir_name**.
 - **cd** : Changes the working directory to your home directory
 - **cd -** : Changes the working directory to the previous working directory
 - **cd ~user_name**: Changes the working directory to the home directory of user_name.

Introduction to Linux - Some bash commands

- ❖ **pushd <dir_name>/ popd** are useful to change the directory and return to where we were.
- ❖ **mkdir directory...** : is used to create directories
 - **-p** makes parent directories as needed. If a directory exists, no error generated
- ❖ **touch file...** : is used to create files. If a file exists, the file timestamp is updated
- ❖ **rm [options] item...** : is used to remove files and directories.
 - Options are **-i**, **-d**, **-l**, **-f**, **-r**, **-v** and combination of them.
- ❖ **cp [options] item1 item2**: copies a single file or directory, **item1**, to file or directory **item2**
- ❖ **cp [options] item... directory**: copies multiple **items** (files or directories) into a **directory**
- ❖ **mv [options] item1 item2**: moves or renames **item1** to **item2**
- ❖ **mv [options] item... directory**: moves one or more items from a directory to another.
- ❖ **cat [file...]**: reads and concatenate one or more files and copies them to the terminal
 - If there is no filename arguments, cat copies standard input to standard output (Ctrl+d)

Introduction to Linux - Some bash commands

- ❖ **echo [options] [string]:** a builtin shell command to print **string** to the standard output.
 - Use **-e** to enable the function of backslash(\) character. E.g. **echo -e "A\nB"** and **echo ***
- ❖ **printf "format" arguments:** a shell builtin to [format and print](#) string to the terminal.
- ❖ **nano** is a simple text editor. E.g. nano text.txt
- ❖ To run multiple commands in one line use semicolon(;). E.g. clear; exit
 - Possible to use logical and(&&) and logical or(||). E.g. touch file && cat file
 - Every program returns an exit code to the system. To get exit code of a program: **echo \$?**
- ❖ **ln file link:** to create a hard link to a file.
- ❖ **ln -s item link:** to create a symbol link to a file or directory, item
- ❖ Access rights to files and directories are defined as
 - Read(**r**) access, write(**w**) access and execution(**x**) access.

Introduction to Linux - Some bash commands

- ❖ A file/directory may be accessible by its **owner**, a **group** of users, or other users(**world**)
- ❖ **chmod permission item...** : to change the mode(permissions) of files or directories
- ❖ **chown [owner][:[group]] file...** : to change owner and group owner of files and directories.
- ❖ In Unix any I/O is a file. It is possible to redirect I/O. For examples
 - **> file.txt**: creates a file or if the file exists, it will be truncated
 - **echo -n "Hello" > file.txt** and **echo " World!" >> file.txt**
 - **cat > file.txt** to copy standard input to file. (press Ctrl+d at the end)
 - **cat < file.txt** : Changes the source from the keyboard to file.txt
- ❖ It is possible to read data from the standard output and send it to the standard input using the pipe operator (|).
 - E.g. **cat file.txt | sort > sort.txt** and **tree /var | less**

drwxrwxrwx

d = Directory
r = Read
w = Write
x = Execute

chmod 777

rwX|rwX|rwX
Owner|Group|Others

7	rwX	111
6	rw-	110
5	r-X	101
4	r--	100
3	-wX	011
2	-w-	010
1	--X	001
0	---	000

Introduction to Linux & Bash

❖ Some useful links

- [Terminal vs. Bash vs. Command line vs. Prompt](#)
- [Linux Architecture and the Kernel](#)
- [Linux kernel architecture and features](#)
- [Architecture of the Linux kernel](#)
- [Anatomy of the Linux kernel](#)
- [Beginner's Guide to Bash](#)
- [Linux/Unix Tutorial](#)